HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2025)

Student No.: 05 Name: _____ Marks: / 98

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. It is given that $PQRS$ is a trapezium with $PQ \parallel SR$. If $PS = 41$ cm, $RS = 53$ cm, $\angle PSR = 120^\circ$ and $\angle QRS = 150^\circ$, then $PQ =$

   A. 82 cm.

   B. 100 cm.

   C. 106 cm.

   D. 135 cm.

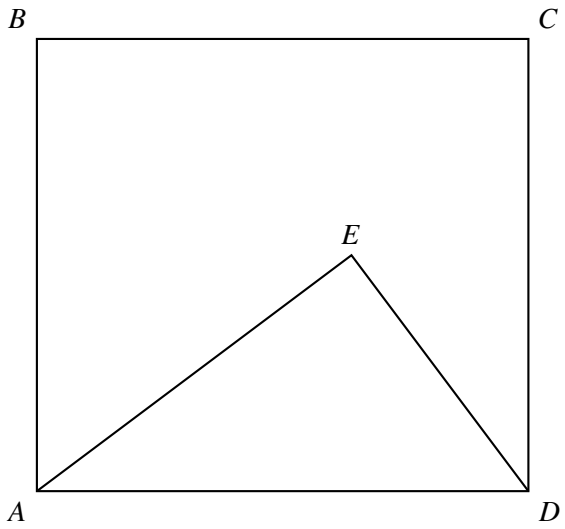
(DSE-2025 PII Q20)

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2. In the figure, $ABCD$ is a rectangle. It is given that $AE = 20\,\text{cm}$ and $\angle AED = 90°$. If the area of $\triangle ADE$ is $150\,\text{cm}^2$, then the perpendicular distance from $E$ to $CD$ is




A. 8 cm.

B. 9 cm.

C. 12 cm.

D. 15 cm.

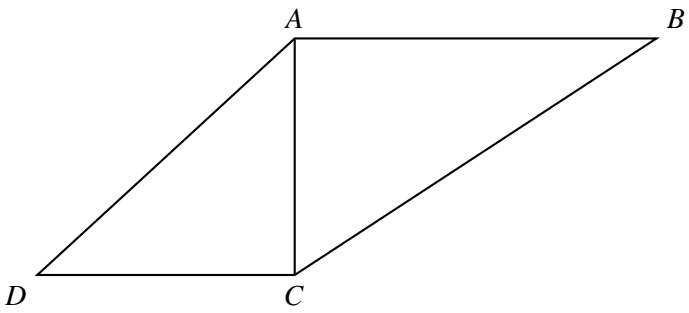
(DSE-2025 PII Q21)

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3. The figure shows the quadrilateral $ABCD$. If $\angle ABC + \angle ADC = \angle ACD = \angle BAC = 90°$, which of the following must be true?




A. $\tan\angle ACB = \dfrac{AB}{AD}$

B. $\tan\angle ACB = \dfrac{AB}{CD}$

C. $\tan\angle ADC = \dfrac{BC}{AD}$

D. $\tan\angle ADC = \dfrac{BC}{CD}$

(DSE-2025 PII Q23)

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4. The polar coordinates of the points $X$, $Y$ and $Z$ are $(1, 20^\circ)$, $(2, 80^\circ)$ and $(r, \theta)$ respectively, where $20^\circ < \theta < 80^\circ$. If $\triangle XYZ$ is an equilateral triangle, find $r$.

A. $\sqrt{3}$

B. $\sqrt{5}$

C. $\sqrt{7}$

D. $\sqrt{10}$

(DSE-2025 PII Q24)

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5. The straight lines $L_1$ and $L_2$ are perpendicular to each other. The equations of $L_1$ and $L_2$ are $3x+4y-20=0$ and $mx+ny-20=0$ respectively, where $m$ and $n$ are constants. Suppose that $L_1$ and $L_2$ cut the $x$-axis at the points $A$ and $B$ respectively. Let $C$ be the point of intersection of $L_1$ and $L_2$. It is given that $C$ lies above the $x$-axis. If the area of $\triangle ABC$ is 6, find $n$.

A. $-12$

B. $-9$

C. 9

D. 12

(DSE-2025 PII Q26)

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6. A box contains six cards numbered 2, 2, 2, 3, 4 and 5 respectively. In a game, a number is randomly drawn from the box and a certain number of tokens will be got according to the following table:

| Number drawn | 2 | 3 | 4 | 5 |
|---|---|---|---|---|
| Number of tokens got | 10 | 15 | 25 | 50 |

Find the expected number of tokens got in the game.

A. 3

B. 4

C. 20

D. 25

(DSE-2025 PII Q28)

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7. $3E000000000000_{16} =$

A. $2^{16}+2^{15}+2^{14}+2^{13}+2^{12}$.

B. $2^{17}+2^{16}+2^{15}+2^{14}+2^{13}$.

C. $2^{52}+2^{51}+2^{50}+2^{49}+2^{48}$.

D. $2^{53}+2^{52}+2^{51}+2^{50}+2^{49}$.

(DSE-2025 PII Q31)

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8. The L.C.M. of $p^2-4q^2$, $p^3-8q^3$ and $(p+2q)(p^2-4q^2)$ is

A. $p-2q$.

B. $p^2-4q^2$.

C. $(p+2q)^2(p^3-8q^3)$.

D. $(p+2q)(p^2-4q^2)(p^3-8q^3)$.

(DSE-2025 PII Q32)

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9. It is given that $\log_5 y$ is a linear function of $\log_{25} x$. The intercepts on the vertical axis and on the horizontal axis of the graph of the linear function are 12 and 2 respectively. If $y = mx^n$, then $n =$

A. $-12$.

B. $-4$.

C. $-3$.

D. $-1$.

(DSE-2025 PII Q33)

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10. On the same rectangular coordinate system, the graph of $y = \log_a x$ cuts the graph of $y = a^x$ and the $x$-axis at the points $P$ and $Q$ respectively, where $a$ is a positive constant. Denote the origin by $O$. Which of the following are true?

I. $a < 1$

II. $OQ > a$

III. $\angle POQ = 45°$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2025 PII Q34)

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11. $i^9 + i^{10} + i^{11} + \cdots + i^{999} =$

A. $-1$.

B. $0$.

C. $1$.

D. $i$.

(DSE-2025 PII Q35)

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12. Consider the following system of inequalities:

$$\begin{cases} x \leq 11 \\ 4x + 5y - 19 \geq 0 \\ 7x - 6y + 11 \leq 0 \end{cases}$$

Let $D$ be the region which represents the solution of the above system of inequalities. If $(x, y)$ is a point lying in $D$, then the greatest value of $8x - 6y + 11$ is

A. $1$.

B. $11$.

C. $15$.

D. $129$.

(DSE-2025 PII Q36)

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13. Let $p$, $q$ and $r$ be non-zero real numbers. If $p$, $q$, $r$ is an arithmetic sequence, which of the following must be true?

I. $3^p$, $3^q$, $3^r$ is a geometric sequence.

II. $\frac{5}{p}$, $\frac{5}{q}$, $\frac{5}{r}$ is a geometric sequence.

III. $p-q$, $q-r$, $r-p$ is an arithmetic sequence.

A. I only

B. II only

C. I and III only

D. II and III only

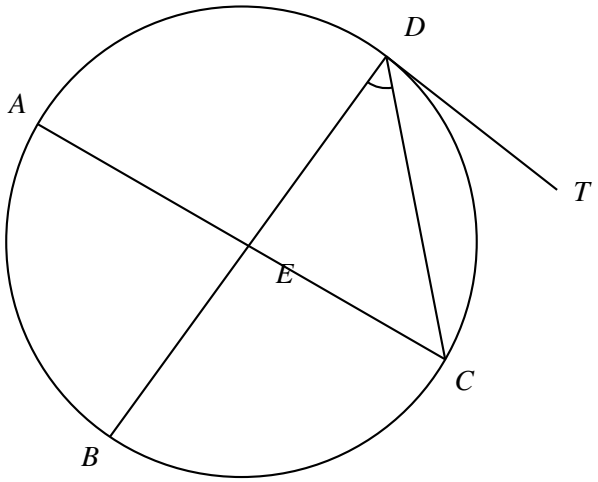
(DSE-2025 PII Q37)

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14. In the figure, $AC$ is a diameter of the circle $ABCD$. Denote the point of intersection of $AC$ and $BD$ by $E$. It is given that $TD$ is the tangent to the circle at $D$. If $\angle BEC = 96^\circ$ and $\angle CDT = 41^\circ$, then $\angle CDE =$




A. $47^\circ$.

B. $48^\circ$.

C. $52^\circ$.

D. $55^\circ$.

(DSE-2025 PII Q38)

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15. $PQRS$ is a regular tetrahedron. Find the angle between $PQ$ and $\triangle QRS$ correct to the nearest degree.

A. 35°

B. 55°

C. 60°

D. 71°

(DSE-2025 PII Q40)

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16. Denote the origin by $O$. The coordinates of the point $U$ are $(20, 0)$. Let $V$ be a point lying on the positive $y$-axis such that the $x$-coordinate of the in-centre of $\triangle OUV$ is 6. Find the area of $\triangle OUV$.

A. 70

B. 87

C. 210

D. 250

(DSE-2025 PII Q41)

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17. There are 9 cans of apple juice and 4 cans of grape juice in a bag. If 6 cans are randomly chosen from the bag at the same time, find the probability that at most 3 cans of grape juice are chosen.

A. $\frac{9}{13}$

B. $\frac{133}{143}$

C. $\frac{140}{143}$

D. $\frac{714}{715}$

(DSE-2025 PII Q43)

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18. The standard scores of a boy and a girl in a Mathematics test are $-2$ and $z$ respectively. The standard deviation of the scores of the Mathematics test is 2 marks. If the difference of the test score of the boy and the test score of the girl is 6 marks, find $z$.

A. $-5$ or 1

B. $-5$ or 3

C. $-3$ or 1

D. $-3$ or 3

(DSE-2025 PII Q44)

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19. It is given that $a$, $b$, $c$ and $d$ are four distinct real numbers. Let $m_1$, $r_1$ and $v_1$ be the mean, the range and the variance of the group of numbers $\{a,b,c,d\}$ respectively while $m_2$, $r_2$ and $v_2$ be the mean, the range and the variance of the group of numbers $\{2a,2b,2c,2d\}$ respectively. Denote the mean, the range and the variance of the group of numbers $\{a+3,b+3,c+3,d+3\}$ by $m_3$, $r_3$ and $v_3$ respectively. Which of the following are true?

I. $m_1+m_3>m_2$

II. $r_1+r_3=r_2$

III. $v_1+v_3<v_2$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2025 PII Q45)

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20. The selling price of a souvenir is $378. The souvenir is sold at a discount of 60% on its marked price. The marked price of the souvenir is 75% above its cost.

(a) Find the marked price of the souvenir.

(b) Find the cost of the souvenir.

(c) Determine whether there is a gain or a loss after selling the souvenir. Explain your answer.

(5 marks) (DSE-2025 PI Q07)

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21. Let $f(x) = 2x^3 + hx^2 + kx + 15$, where $h$ and $k$ are constants. When $f(x)$ is divided by $x+2$, the remainder is $-45$. It is given that $f(x)$ is divisible by $2x-5$.

(a) Find $h$ and $k$.

(b) Someone claims that all the roots of the equation $f(x) = 0$ are rational numbers. Do you agree? Explain your answer.

(6 marks) (DSE-2025 PI Q10)

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22. It is given that $p(x)$ partly varies as $x$ and partly varies as $x^2$. Suppose that $p(7) = 56$ and $p(9) = 216$.

(a) Find $p(x)$.

(b) Let $c$ be a real constant. Find the range of values of $c$ such that the equation $p(x) = c$ has two distinct real roots.

(6 marks) (DSE-2025 PI Q11)

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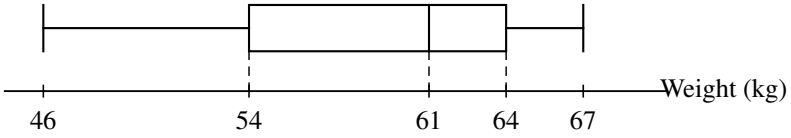
23. The stem-and-leaf diagram below shows the distribution of the weights (in kg) of some athletes before training.

| Stem (tens) | Leaf (units) |
|---|---|
| 4 | 5 6 8 |
| 5 | $w$ $w$ 6 7 |
| 6 | 3 4 5 6 9 9 |
| 7 | 0 2 |
| 8 | 7 |

The difference of the range and the inter-quartile range of the above distribution is 25 kg.

(a) Write down the range of the above distribution. Hence, find $w$.

(b) The box-and-whisker diagram below shows the distribution of the weights (in kg) of the athletes after the training.




(i) Find the change in the upper quartile of the distribution due to the training.

(ii) Is the distribution of the weights of the athletes after the training less dispersed than that before the training? Explain your answer.

(7 marks) (DSE-2025 PI Q12)

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24. The height and the base radius of a solid metal right circular cone are 45 cm and 24 cm respectively. The circular cone is divided into a circular cone and a frustum $X$ by a plane parallel to its base. It is given that the height of $X$ is 30 cm.

(a) Express the volume of $X$ in terms of $\pi$.

(b) Express the total surface area of $X$ in terms of $\pi$.

(c) $X$ is melted and recast into a solid cube. Does the total surface area of the cube exceed the total surface area of $X$? Explain your answer.

(9 marks) (DSE-2025 PI Q14)

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25. Let $T(n)$ be the $n$th term of an arithmetic sequence such that $T(1) \neq T(2)$ and $T(47) = 456$. It is given that $T(9)$, $T(47)$, $T(199)$ is a geometric sequence.

(a) Find $T(1)$.

(b) Find the least value of $n$ such that the sum of the first $n$ terms of the arithmetic sequence is greater than $10^6$.

(6 marks) (DSE-2025 PI Q17)

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26. Let $g(x) = 3x^2 - 6kx + 24x + 3k^2 - 24k + 55$, where $k$ is a positive constant. Denote the vertex of the graph of $y = g(x)$ by $R$.

(a) Using the method of completing the square, express the coordinates of $R$ in terms of $k$.

(b) The graph of $y = h(x)$ is obtained by translating the graph of $y = g(x)$ rightwards by 6 units and then downwards by 10 units. Denote the vertex of the graph of $y = h(x)$ by $S$. Let $T$ be a point such that the orthocentre of $\triangle RST$ is the origin.

(i) Express the coordinates of $T$ in terms of $k$.

(ii) The coordinates of the point $U$ are $(-5, 5)$. Let $V$ be a point lying on $RS$ such that $TV$ is perpendicular to $RS$. It is given that $S$, $T$, $U$ and $V$ are concyclic. Find $k$.

(9 marks) (DSE-2025 PI Q18)

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27. The equation of the straight line $L$ is $4x-3y+83=0$. The equation of the circle $C$ is $x^2+y^2+ax-2y+b=0$, where $a$ and $b$ are positive constants. It is given that $C$ passes through the point $(-10,9)$ and $L$ is a tangent to $C$.

(a) Prove that the equation $25x^2+(9a+640)x+(90a+4924)=0$ has equal roots. Hence, find $a$ and $b$.

(b) The coordinates of the point $P$ are $(-20,1)$. Let $Q$ be a point lying below $L$ such that $PQ$ is tangent to $C$ and $PQ=25$. Denote the centre of $C$ by $I$.

(i) Find $\cos\angle IPQ$.

(ii) Find, in surd form, the distance between $I$ and $Q$.

(iii) Let $R$ be a point lying on $L$ such that $C$ is the inscribed circle of $\triangle PQR$. Denote the radius of the circumcircle of $\triangle PQR$ by $r$. Someone claims that $r>PQ$. Is the claim correct? Explain your answer.

(12 marks) (DSE-2025 PI Q19)

I can finish _____ questions

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